Canadian Life Insurance Planning & Pricing Platform

Project Report — Educational Actuarial Analytics Portfolio Project

■■ ***Educational student project.*** *All figures, product names, and premium estimates in this report are illustrative and do not constitute real insurance quotes, underwriting decisions, or professional financial advice. Please consult a licensed insurance advisor before making coverage decisions.*

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Live Demo	canadian-life-insurance-platform.streamlit.app
Repository	github.com/AliceChen29/canadian-life-insurance-platform
Tech Stack	Python, Streamlit, R

1. Executive Summary

This project is an educational web platform that helps individuals understand their life insurance planning options before speaking with a licensed advisor. It combines five interactive tools — an insurance needs calculator, a product recommendation engine, a risk scoring model, an illustrative premium estimator, and an investment growth projector — with a statistical analysis layer built in R.

The platform is deliberately positioned as an **educational tool**, not a real insurance quoting or underwriting system. This distinction matters both practically (insurance advice is a regulated activity in Canada) and pedagogically: the project's value lies in demonstrating actuarial and statistical reasoning, not in producing production-grade insurance products.

A key statistical finding is that a Gamma Generalized Linear Model (GLM) reduces Mean Absolute Error by approximately 14.5% compared to an ordinary least squares baseline when predicting illustrative premiums — consistent with the theoretical expectation that premium data, being strictly positive and right-skewed, is better modeled by a Gamma distribution than by a normal-error linear model.

2. Business Problem & Target User

Many Canadians find it difficult to estimate how much life insurance coverage they actually need, or to understand how factors like age, smoking status, and health risk translate into cost. This often leads people to either avoid the topic entirely or rely solely on the recommendation of a single advisor without an independent frame of reference.

This platform targets individuals who want to build baseline understanding and prepare informed questions **before** meeting with a licensed insurance advisor. It is not intended to replace that advisor relationship.

3. Project Scope

The first version of the platform intentionally focuses on individual life insurance only (no auto, health, or home insurance), and covers five modules:

- **Insurance Needs Calculator** — estimates required coverage using an income replacement method with present-value discounting.
- **Product Recommendation** — scores four illustrative product types (Term 10 / 20 / 30, Permanent Life) against user inputs.
- **Risk Score** — a 0–100 illustrative risk score based on age, smoking, health, and occupation factors.
- **Premium Estimation** — an illustrative premium formula combining base rate, age, smoking, health, and term factors.
- **Investment Projection** — deterministic and Monte Carlo projections of long-term investment growth.

4. Methodology

4.1 Insurance Needs Calculation

Coverage need is estimated using an income replacement approach with two configurable assumptions: a **replacement rate** (default 75%, reflecting that household expenses typically fall when one income is lost, rather than assuming 100% replacement) and a **discount rate** (default 3%) applied via the present value of an ordinary annuity:

$$PV = (\text{Annual Income} \times \text{Replacement Rate}) \times [1 - (1 + r)^{-n}] / r$$

Total coverage need = PV of income replacement + outstanding debt + education need + final expenses – existing savings – existing coverage. Both assumptions are exposed as adjustable sliders in the application so users can see how sensitive the result is to each assumption.

4.2 Risk Scoring Model

An illustrative 0–100 risk score is computed from age, smoking status, BMI category, pre-existing medical conditions, and occupation risk, then mapped to Low / Moderate / High / Very High classifications. This is explicitly not a real underwriting model.

4.3 Product Recommendation

Four illustrative product types are scored using a weighted rule-based system considering desired term length, number of dependants, budget level, and estate planning needs. The product with the highest score is recommended, with the full score breakdown shown to the user for transparency.

4.4 Premium Estimation

Annual premium = Coverage × Base Rate × Age Factor × Smoking Factor × Health Factor × Term Factor. All factors are illustrative multipliers documented in the accompanying Excel assumption workbook, not real actuarial mortality tables.

4.5 Investment Projection

A deterministic future value calculation (compound interest with regular contributions) is paired with a Monte Carlo simulation (1,000 paths) that samples monthly returns from a normal distribution to show a realistic range of outcomes (10th / 50th / 90th percentile) rather than a single deterministic figure.

5. Statistical Analysis

A synthetic dataset of 3,000 applicant records was generated for exploratory data analysis and model comparison in R. No real personal data was used at any point in this project.

5.1 Exploratory Data Analysis

Distributions of age, premium, and their relationship to smoking status were examined using ggplot2. Estimated premiums were confirmed to be right-skewed and strictly positive, motivating the choice of a Gamma-distributed response model over ordinary least squares.

5.2 Model Comparison

Two models were fit to predict estimated annual premium: an OLS linear regression baseline, and a Gamma GLM with a log link function.

Model	MAE ($)	RMSE ($)
Linear Regression	687.04	1,023.85
Gamma GLM	587.27	989.23

The Gamma GLM reduced Mean Absolute Error by approximately **14.5%** relative to the linear baseline, consistent with theoretical expectations for positively-skewed, strictly-positive cost data.

5.3 Model Coefficients (Gamma GLM)

Term	Estimate	Std. Error	t value
(Intercept)	4.659	0.0287	162.30
age	0.0282	0.0005	61.07
smoker (Yes)	0.6914	0.0185	37.46
coverage	0.000001	0.00000001	114.14
term_years	0.0093	0.0008	11.83

All coefficients are statistically significant at conventional levels. On the log scale, smoking status is associated with the largest single effect on estimated premium among the categorical predictors.

6. Platform Overview

The platform is built with Python and Streamlit, deployed on Streamlit Community Cloud, and structured as a multi-page application: a landing screen with a clear educational disclaimer, followed by five tool pages (Needs Calculator, Recommendation, Premium Calculator, Investment Projection, and Model Insights). The Model Insights page surfaces the R statistical analysis results directly inside the web application.

7. Limitations

• All applicant data used for statistical analysis is synthetically generated, not real.

• Premium, risk score, and coverage figures are illustrative and do not reflect real actuarial mortality tables, real insurer pricing, or real underwriting practices.

• The model does not account for adverse selection, lapse rates, or policy expenses beyond the simplified loadings described above.

• This tool is not a substitute for advice from a licensed insurance advisor and does not constitute financial advice.

8. Future Improvements

• Incorporate real (published, anonymized) mortality table structures for the premium model, clearly cited and still framed as educational.

• Add user accounts to save and compare multiple scenarios over time.

• Extend the recommendation engine with a machine learning classifier trained on a larger synthetic dataset, benchmarked against the current rule-based system.

• Add French-language support, given the bilingual nature of the Canadian insurance market.

9. Conclusion

This project demonstrates an end-to-end actuarial analytics workflow: business problem framing, transparent rule-based modeling, statistical model comparison in R, and deployment as an interactive web application. The explicit framing as an educational tool — rather than a real insurance product — reflects an understanding of the regulatory boundaries around insurance advice in Canada, while still allowing the underlying actuarial and statistical reasoning to be demonstrated in full.

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